

Week 6HW R functions

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Table of contents

(A. Can you improve this analysis code?)	1
(B. Can you improve this analysis code?)	2
Q1. What type of object is returned from the read.pdb() function?	5
Q2. What does the trim.pdb() function do?	5
Q3. What input parameter would turn off the marginal black and grey rectangles in the plots and what do they represent in this case?	5
Q4. What would be a better plot to compare across the different proteins?	5
Q5. Which proteins are more similar to each other in their B-factor trends. How could you quantify this?	5
Q6. How would you generalize the original code above to work with any set of input protein structures?	6

#Section 1: Improving analysis code by writing functions

(A. Can you improve this analysis code?)

Function input: a numeric vector It rescales values to range from 0 to 1 Function output: a rescaled numeric vector

```
rescale01 <- function(x) {  
  if(all(is.na(x))) {  
    return(x)  
  }  
  (x-min(x, na.rm = TRUE)) / (max(x, na.rm = TRUE) - min(x, na.rm = TRUE))  
}  
df <- data.frame(
```

```

a=1:10,
b=seq(200,400,length=10),
c=11:20,
d=NA
)

df[] <- lapply(df, rescale01)
print(df)

```

	a	b	c	d
1	0.0000000	0.0000000	0.0000000	NA
2	0.1111111	0.1111111	0.1111111	NA
3	0.2222222	0.2222222	0.2222222	NA
4	0.3333333	0.3333333	0.3333333	NA
5	0.4444444	0.4444444	0.4444444	NA
6	0.5555556	0.5555556	0.5555556	NA
7	0.6666667	0.6666667	0.6666667	NA
8	0.7777778	0.7777778	0.7777778	NA
9	0.8888889	0.8888889	0.8888889	NA
10	1.0000000	1.0000000	1.0000000	NA

(B. Can you improve this analysis code?)

```

library(bio3d)

plot_protein_bfactor <- function(pdb_id, chain_id = "A") {
  pdb <- read.pdb(pdb_id)
  pdb_chain <- trim.pdb(pdb, chain = chain_id, eley = "CA")
  bfactor <- pdb_chain$atom$b

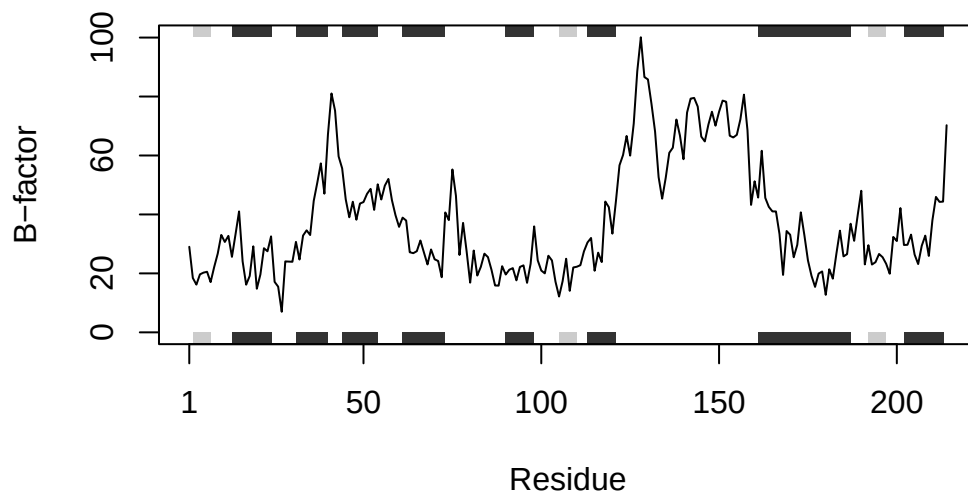
  plotb3(bfactor, sse = pdb_chain, typ = "l", ylab = "B-factor")

  return(bfactor)
}

```

```
s1 <- plot_protein_bfactor("4AKE") # kinase with drug
```

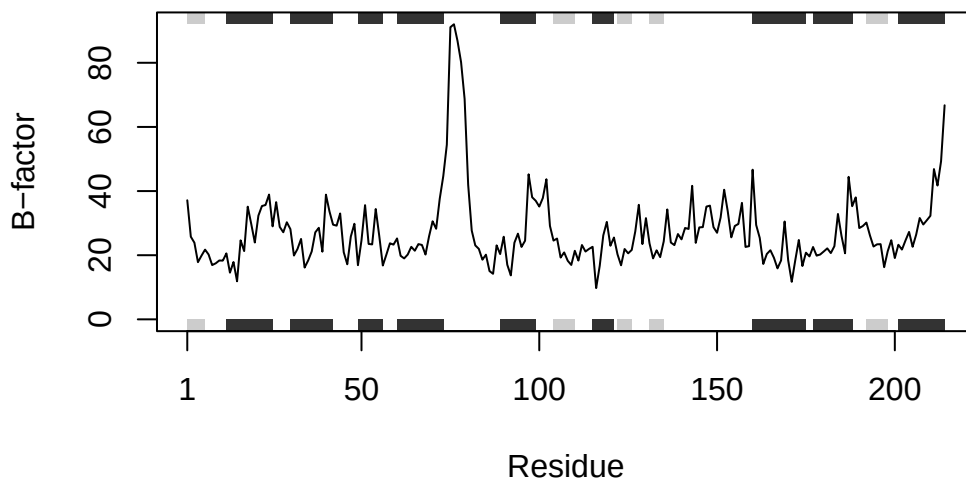
Note: Accessing on-line PDB file



```
s2 <- plot_protein_bfactor("1AKE") # kinase no drug
```

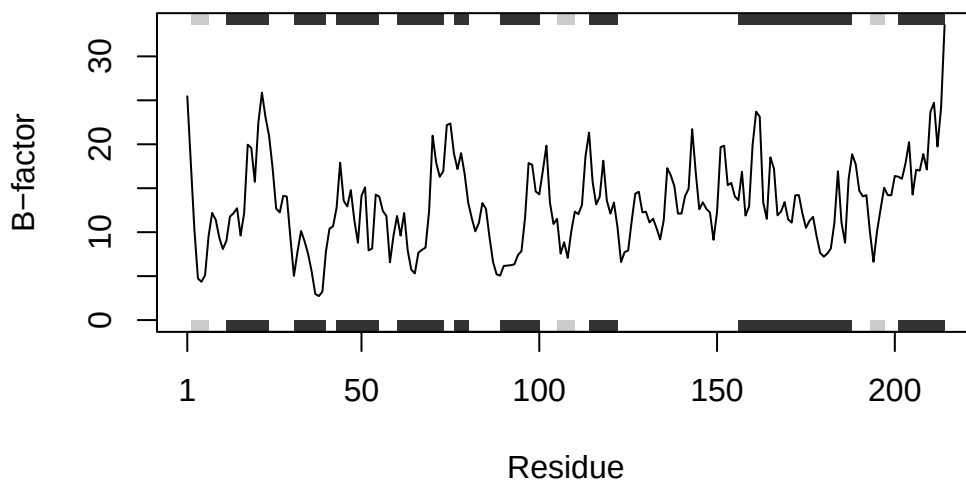
Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE



```
s3 <- plot_protein_bfactor("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file



Inputs: `pdb_id` = PDB identifier `chain_id` = protein chain to analyze

What it does: It reads the PDB structure. It also keeps only the alpha carbons from the chosen chain, extracts B-factors and then plots them.

Output: A B-factor plot and numeric B-factor vector is returned invisibly

Q1. What type of object is returned from the `read.pdb()` function?

The function returns a PDB object in R. The PDB object contains structural information from the protein file.

Q2. What does the `trim.pdb()` function do?

The function `trim.pdb()` subsets the PDB structure to only keep the selected atoms or residues. For this example, it kept Chain A and atoms of type CA.

Q3. What input parameter would turn off the marginal black and grey rectangles in the plots and what do they represent in this case?

The `sse = FALSE` turns off the marginal black and grey rectangles. They represent the secondary structure elements for the protein.

Q4. What would be a better plot to compare across the different proteins?

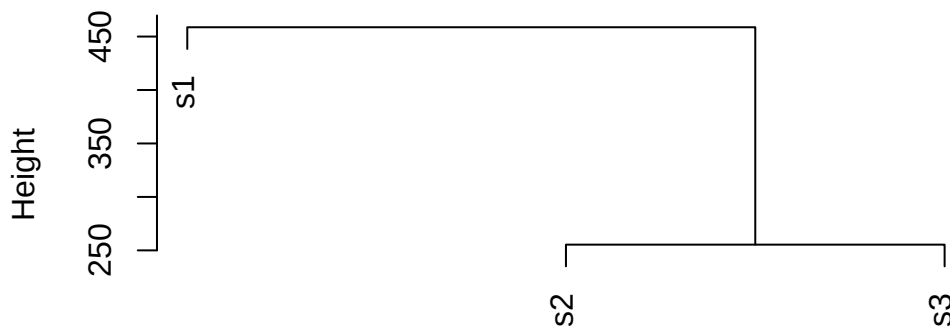
A single overlay line plot

Q5. Which proteins are more similar to each other in their B-factor trends. How could you quantify this?

The two proteins that cluster together in the dendrogram are more similar to each other in their B-factor trends. I can quantify this by using `rbind()` to combine the B-factor vectors, `dist()` to calculate the distances, and `hclust()` clustering.

```
hc <- hclust( dist( rbind(s1, s2, s3) ) )  
plot(hc)
```

Cluster Dendrogram



```
dist(rbind(s1, s2, s3))  
hclust (*, "complete")
```

Q6. How would you generalize the original code above to work with any set of input protein structures?

```
library(bio3d)  
  
# Function inputs:  
# pdb_id = 4-character Protein Data Bank ID  
# chain_id = protein chain to analyze  
  
# What function does:  
# The function reads a PDB structure, trims it to the selected chain and CA atoms, extracts t  
  
# How to use:  
# Use the analyze_protein("4AKE") function  
# Function output:  
# It returns the B-factor values for selected chain as a numeric vector  
  
analyze_protein <- function(pdb_id, chain_id = "A") {  
  #Read the PDB structure  
  pdb <- read.pdb(pdb_id)
```

```

#Keep chose chain and CA atoms
pdb_chain <- trim.pdb(pdb, chain = chain_id, eley = "CA")

#Extract B-factors
b_factors <- pdb_chain$atom$b

#Plot B-factors
plotb3(b_factors,
      sse = FALSE,
      typ = "l",
      ylab = "B-factor",
      xlab = "Residue Position",
      main = paste("B-factor plot for", pdb_id, "chain", chain_id))

#Return B-factor vector
return(b_factors)
}

#Function calls
s1 <- analyze_protein("4AKE")

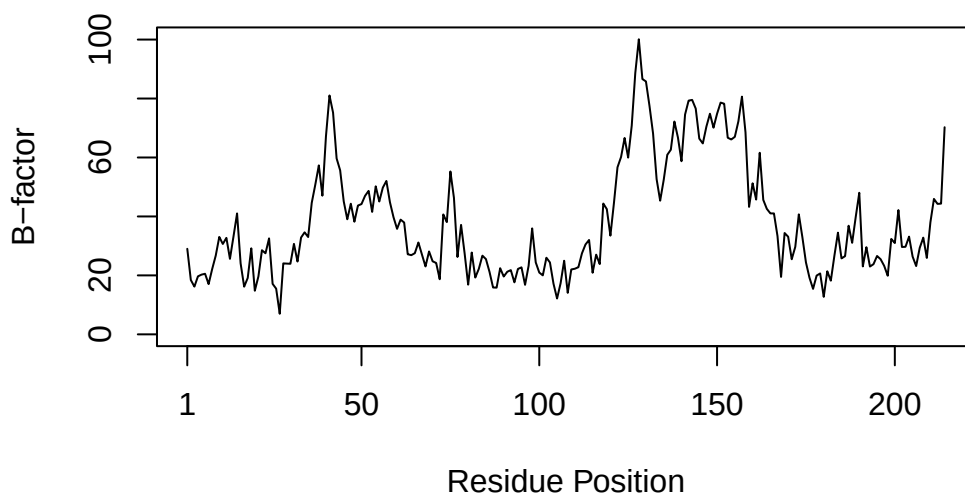
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
 /var/folders/31/df3xvnjs2p1369hdbggqpzbm0000gn/T//RtmpkvG0Gi/4AKE.pdb exists.
 Skipping download

Warning in plotb3(b_factors, sse = FALSE, typ = "l", ylab = "B-factor", :
 Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

B-factor plot for 4AKE chain A



```
s2<- analyze_protein("1AKE")
```

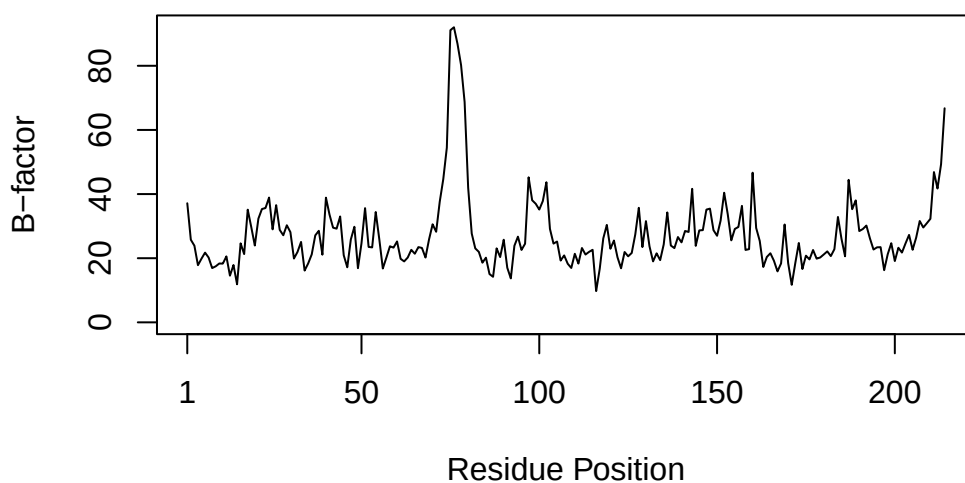
Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/31/df3xvnjs2p1369hdbggqpzbm0000gn/T/RtmpkvGOGi/1AKE.pdb exists.
Skipping download

PDB has ALT records, taking A only, rm.alt=TRUE

Warning in plotb3(b_factors, sse = FALSE, typ = "l", ylab = "B-factor", :
Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

B-factor plot for 1AKE chain A



```
s3 <- analyze_protein("1E4Y")
```

Note: Accessing on-line PDB file

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):  
/var/folders/31/df3xvnjs2p1369hdbggqpzbn0000gn/T/RtmpkvG0Gi/1E4Y.pdb exists.  
Skipping download  
Warning in get.pdb(file, path = tempdir(), verbose = FALSE): Length of input  
'sse' does not equal the length of input 'x'; Ignoring 'sse'
```

B-factor plot for 1E4Y chain A

